

SSZ P5 / Horndeski-SVT

Finite-I Closure and QNM Threshold

Detailed technical memory and reproducibility report

Frozen status after the final variable-G4 outer candidate audit
16 September 2026

Purpose: external technical memory for later reconstruction by Carmen and ChatGPT.
This is not a publication claim; it records what was proved, what was merely constructive, what was superseded, and exactly where the final explicit representative failed.

STOP CHECKPOINT

Do not restart ad-hoc representative hunting from scratch. The 41/41 coefficient language, derivative service, constraint maps, Zhang-Kase emitter, Maxwell-Horndeski emitter, and profile reducer are already closed. The unresolved task is a targeted global finite-I representative-selection/control problem.

1. Executive status - the short version

The most important distinction in the whole project is that several different meanings of "closure" were used at different stages. The apparent contradiction between "41/41 closed" and the final finite-I failure disappears once these levels are separated.

Object	Status	Meaning
P5 frozen geometry	PASS	Regular, horizonless, two light rings resolved.
Local sector principal witnesses	PASS	Exterior Horndeski, central genuine-SVT, regular Horndeski core.
Constructive local same-action handovers	PASS	C-infinity existence with background re-solving and transverse jet controls.
Unreduced 41-slot coefficient language	CLOSED	Complete a,b,c,d,e,v tables for selected local/sector representatives.
JET9D8 radial derivative service	PASS	Needed higher radial jets and product rules regress.
Full algebraic even constraint maps	PASS	All common pivots nonzero on certified domains.
Profile-aware K,R,G,S,M reducer	PASS	Includes product-rule/IBP terms and antisymmetric S block.
Zhang-Kase Appendix-A emitter	PASS	Complete 41-slot emitter, regression-controlled.
Variable-G4(phi) Maxwell-Horndeski emitter	PASS	Strong-H 39/41 regression after c6,d3,e4 parser fixes.
Global canonical principal K/R/G stream	PASS as principal witness	45,617-point center-to-infinity characteristic stream.
Variable-G4 outer finite-I candidate	REJECTED	sym(K) is negative over much of the band; moreover the variable-G4 transplant was not followed by a fresh full background re-solve.
Certified global finite-I same-action member	OPEN	No center-to-infinity on-shell member has yet passed the complete finite-I gates.
Coupled QNM spectrum	BLOCKED	Must not be computed/claimed until a certified global finite-I member exists.

Bottom line

The statement "41/41 is closed" remains correct. The Sep-16 variable-G4 outer construction is a rejected diagnostic candidate, not a certified global same-action member: after importing the healthier Horndeski principal carrier, the complete background equations were not re-solved for that modified member. Its negative sym(K) is enough to reject the candidate, but it is not a no-go for the still-open global finite-I same-action problem.

1.1 Kinetic result of the final variable-G4 candidate

L	min eig(K)	negative K rows	min radial c^2	neg c^2	min Dh1	min DeltaV	min 2Lv9
6	-2116.85	2113 / 2201	0.0888293	0	3.3874	0.187126	12.8789
12	-668.848	2065 / 2201	0.0906121	0	14.1102	0.187126	25.7579
20	-236.372	2025 / 2201	0.0638748	0	39.8178	0.187126	42.9298
42	-12.8644	1903 / 2201	0.00285261	0	177.755	0.187126	90.1525
110	-1.88458	1576 / 2201	0.000950391	0	1227.61	0.187126	236.114
420	-0.250559	1061 / 2201	0.000254172	0	17952.1	0.187126	901.525
1000	-0.10648	629 / 2201	0.000107519	0	101834	0.187126	2146.49

Where K is positive, the radial generalized eigenvalues in this audit remain positive. That does not rescue a ghost-sign failure of K itself. The kinetic gate is logically prior to a physical QNM interpretation.

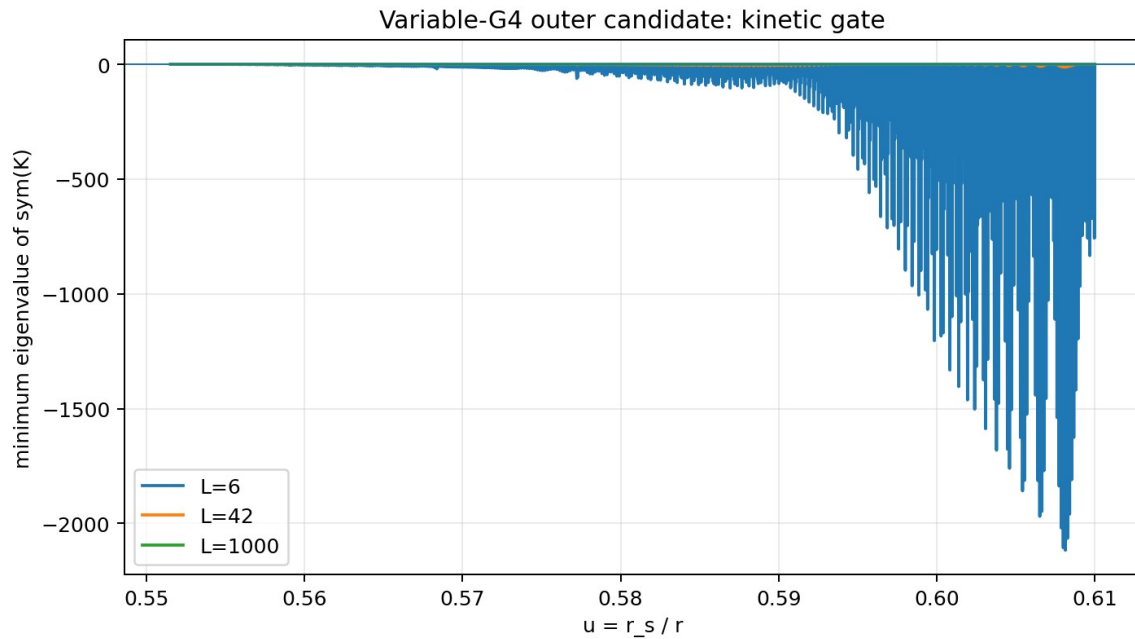


Figure 1. Minimum eigenvalue of $\text{sym}(K)$ for the final variable-G4 outer candidate, $L=6, 42$ and 1000 .

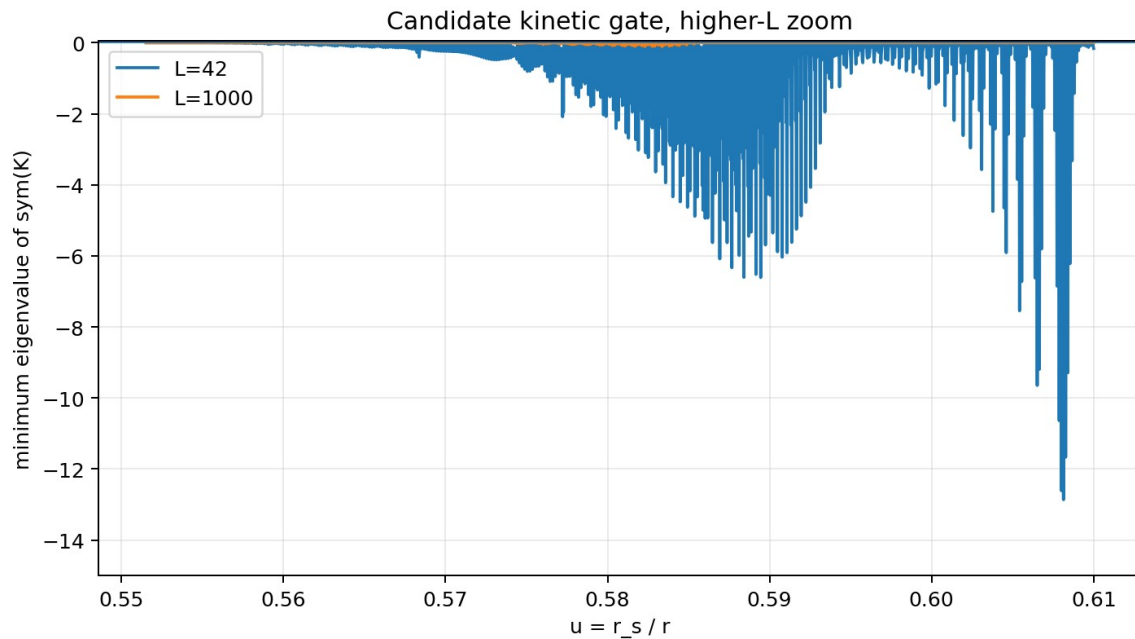


Figure 2. Higher-L zoom of the rejected candidate. The negative direction weakens with increasing L but persists.

2. Terminology: what exactly was meant by "closed"

The project accumulated several distinct closure layers. Keeping them separate is essential for later work.

2.1 Slot closure: 41/41

The common unreduced even-parity coefficient language contains exactly 41 coefficient slots:

$$a_1 \dots a_9 \ (9) + b_1 \dots b_5 \ (5) + c_1 \dots c_6 \ (6) + d_1 \dots d_4 \ (4) + e_1 \dots e_4 \ (4) + v_1 \dots v_{13} \ (13) = 41.$$

"41/41 closed" means that complete unreduced coefficient tables exist for selected local/sector representatives in this common basis. It does not by itself prove that all radial sectors belong to one identical holonomic action member, nor that the reduced finite-I kinetic matrix is positive everywhere.

2.2 Constructive same-action closure

This is an existence statement. The outer and inner overlap bands possess enough background-constrained and background-null action-jet freedom to connect the local representatives by smooth C-infinity handovers while retaining the frozen P5 geometry and nonzero common constraint pivots.

2.3 Principal global closure

A separate center-to-infinity characteristic/principal stream exists. It proves that one can select a globally healthy principal-symbol representation. It is not automatically the same basis, same lower-order member, or same action representative as an independently built finite-I mass/potential block.

2.4 Finite-I global same-action closure

This is the strongest level required before a coupled QNM spectrum can be called physical: one holonomic center-to-infinity action member, with both handovers directly evaluated from that member, reduced with all profile derivatives retained, and passing $K>0$ plus radial and angular gradient gates at finite I. This final level is not yet passed.

3. Frozen P5 geometry and theory architecture

3.1 Background variables

The P5 geometry is source-locked in the segment variable ξ . With $r_s=1$, $C=r_s/r=1/x$ and $\phi=\xi(C)$. The susceptibility $\chi=d\xi/dC$ obeys the autonomous P5 closure $d\xi/dC=(1-\xi)^{3/2} P5(\xi)$. The temporal metric is $f=(1+\xi)^{-2}$; h is fixed by the P5 radial closure.

- Outer unstable light ring: $C=2/3$, $x=1.5$.
- Inner stable light ring: C approximately 0.706135, x approximately 1.41616064.
- Regular horizonless center.

The historical Hermite middle bridge is provenance only and is not the P5 field equation.

3.2 Exact kappa identity and the 00/11 basis issue

$$\phi_r = -C^2 \chi, \quad \kappa = h \phi_r^2 = h C^4 \chi^2 = -2 X.$$

For the metric background block $(f_2, f_2 X)$, the E00/E11 Jacobian has the triangular form $J=[[-A, 0], [-A, -B]]$ with $A=r^2 f$ and $B=A \kappa$. Merely swapping E00 and E11 cannot improve singular values. The algebraically natural basis is instead $E0=E00$ and $E\Delta=E11-E00$, in which the Jacobian is diagonal: $\text{diag}(-A, -A \kappa)$.

This is the precise version of the earlier 00/11 rotation/inversion intuition: the useful operation is not a blind row swap but a regime-aware basis change. The same $\kappa=-2X$ also controls the rank of the simple SVT Hessian control map. Near the exact center $\kappa \rightarrow 0$, so the scalar coordinate chart must be replaced by the separate regular Taylor-center branch rather than divided through by κ .

3.3 Hybrid covariant architecture

$$C_{\text{HSVT}} = C_{\text{MH}} + (C_{\text{SVT}} - C_{\text{shared}}).$$

This addition must be performed on the common unreduced quadratic action before eliminating $H0, H1, H2, h1, dA0, dA1$ or the auxiliary vector variable V . Constraint elimination is nonlinear in the coefficient set, so $\text{Reduce}(A+B)$ is not equal to $\text{Reduce}(A) + \text{Reduce}(B)$ in general.

The physical architecture is therefore Horndeski exterior $\rightarrow$ genuine $U(1)$ -SVT light-ring lobe $\rightarrow$ Horndeski core, with the genuine-SVT lobe containing both light rings.

4. Established local principal witnesses and the global canonical principal stream

4.1 Central genuine-SVT reference

The central exact Zhang-Kase reference on $0.61 \leq u \leq 0.715$ is the strongest local even-sector witness. Regressed characteristic minima include:

Quantity	Certified minimum / range
K_{even}	about 10.5415 > 0
c_r, T^2	≥ 1.00507

Quantity	Certified minimum / range
c_r, S^2	≥ 0.903745
c_r, V^2	≥ 9.45887
$c_{\Omega}, -^2$	≥ 84.912
$c_{\Omega}, +^2$	≥ 643.142

4.2 Global principal export

Region	Rows
Horndeski_core_to_center	29790
central_exact_SVT	3428
central_inner_matching_buffer	401
central_outer_matching_buffer	400
inner_SVT_H_characteristic_handover	1401
outer_H_SVT_characteristic_handover	2201
weak_exterior_H	7996

The 45,617-point principal export has $\min f=0.25$, $\min h=0.326856666$, $\min cT^2=0.9992624$, $\min cS^2=0.903745178$, $\min cV^2=1$. All stored R_{ij} are exactly zero in this canonical export.

Important limitation

This global K/R/G file is a principal-characteristic witness. It must not be silently spliced to an independently constructed finite-I M block and relabeled as a single same-action physical KRGH operator.

5. Same-action handovers and background re-solving

5.1 Outer handover

The explicit re-solved outer background band covers approximately $u=0.551523087$ to 0.61 . The flat partition is genuinely flat at the Horndeski endpoint: $S_{\text{SVT}}=0$ through the initial plateau, rises across the overlap, and reaches $S_{\text{SVT}}=1$ at $u=0.61$. Typical values are approximately $S=0.001$ near $u=0.575$, $S=0.065$ near 0.58 , $S=0.501$ near 0.59 , $S=0.935$ near 0.60 , and 1 at 0.61 .

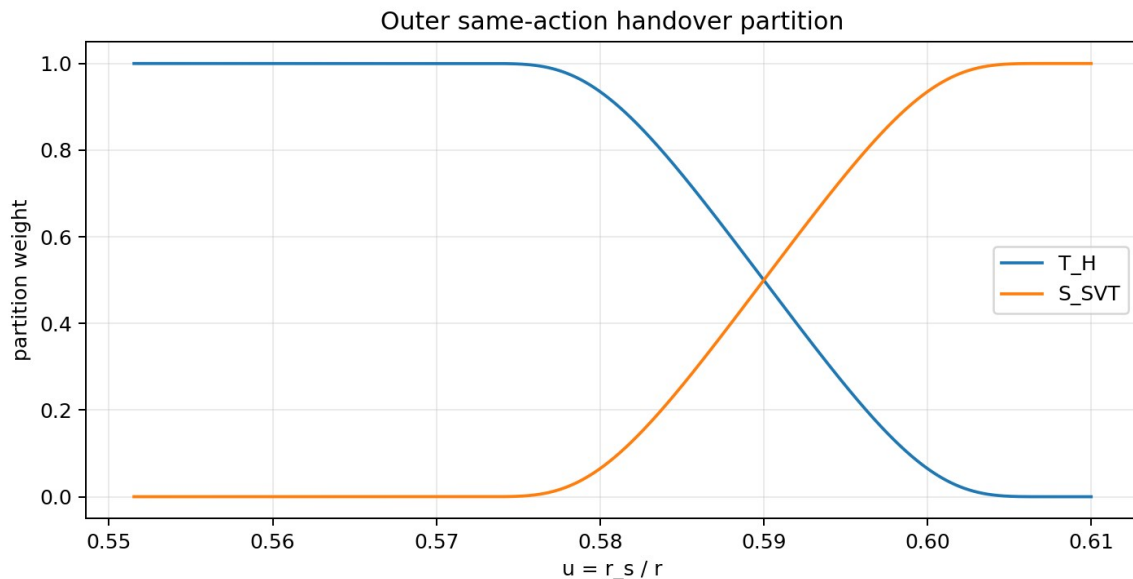


Figure 3. Stored outer C-infinity partition weights T_H and S_{SVT} .

The resolved metric/vector residuals are at machine precision: max |E00_resolved| about 1.8e-16, max |E11_resolved| about 3.2e-16, max |J_A| about 1.6e-15.

5.2 Inner handover

The explicit inner band covers $u=0.71\dots0.715$. The actual re-solved residuals are machine zero (E00 about 8.9e-16, E11 about 4.7e-15, J_A about 3.1e-15). A previous diagnostic column E_H+E_{SVTg} is not the complete residual in this band because partition derivatives and re-solved lower Horndeski jets contribute. Treating that naive sum as a failure was superseded.

The stored genuine-SVT lower values themselves visibly follow the action partition: $f2=S_{SVT}*f2_{full}$ and $f2X=S_{SVT}*f2X_{full}$ to numerical precision, with $f2F$ approximately S_{SVT} . This is why a split block must not be reinterpreted as an unsplit full-SVT background action.

6. Background-null controls and why local existence survived the handovers

The static background fixes only action data tangent to the one-dimensional background trajectory. Transverse jets remain available for perturbative control. For the genuine-SVT Hessian controls $H=(HXX,HXF,HXY,HFF,HFY,HYY)$, the three radial channels are:

$$\begin{aligned}\Delta v1 &= HFF - 4 \kappa HFY + 4 \kappa^2 HYY \\ \Delta v4 &= -HXF + 2 \kappa HXY - 4 Fbg HFY + 8 Fbg \kappa HYY \\ \Delta c2 &= (\kappa/2) HXX - Fbg HXF + 6 Fbg \kappa HXY - 4 Fbg^2 HFY + 16 Fbg^2 \kappa HYY\end{aligned}$$

In the selector gauge $HXY=HFY=HYY=0$, the exact right inverse is:

$$HFF = \Delta v1, \quad HXF = -\Delta v4, \quad HXX = (2/\kappa)(\Delta c2 - Fbg \Delta v4).$$

$$\det M_{rad} = \kappa/2.$$

Certified κ minima are approximately 0.098838 in the outer overlap, 0.207862 in the central lobe, and 0.632753 in the inner overlap, so this control map does not lose rank there. Independent unused Hessian directions remain available for angular-channel regulation. The Horndeski principal-control map with targets $(F,H,a1,c4,c2)$ was likewise found full rank through the onset-to-core interval.

7. Radial derivative service, holonomicity, and branch corrections

7.1 JET9D8

The accepted radial derivative service is a local polynomial jet on the nonuniform $r=x$ grid with window=9 and polynomial degree=8. This replaced spline-based derivative experiments for production finite-I work.

- $v12=-v6/(2h)$ regression: absolute residual around 1e-14 or better.
- $v13$: median scaled relative error around 6e-6 on the central regression; edge maxima are derivative/rounding sensitive.
- $e4$: isolated median at the few-1e-7 scale; end-to-end regression is also strong.

The previously selected plus branch $v12=+v6/(2h)$ is superseded. Pure Horndeski has $v6=0$, so the sign was invisible there, but it matters in genuine-SVT and hybrid bands.

7.2 Correct a5

$$a5 = a2' - a1'' - (A0' v4 / 2)' + A0' v5 / 2.$$

For pure Horndeski $A0'=0$, this reduces to $a5=a2'-a1''$. A previous builder omitted $-a1''$ and is not authoritative.

7.3 Holonomic on-curve partials

Two Appendix expressions exposed the same practical point: in the selected one-dimensional action representative, partial derivatives appearing in $d3$ are represented by the holonomic on-curve derivatives, e.g. $\text{partial_phi } v6=(dv6/dr)/\text{phi_r}$ and $\text{partial_phi } a4=a4'/\text{phi_r}$. Using naive explicit partials generated false mismatches. With the holonomic convention, $d3$ regression closes to numerical/JET accuracy.

7.4 Central action holonomicity

Direct chain-rule checks such as $df3/dr=f3_phi \text{ phi_r}+f3_X X_r$ and the corresponding relations for $f3_X$, $f4$, $f4_X$, $f4_XX$, $f2$ and $f2_X$ close in the interior, usually around 1e-6 or better and much tighter for several upper jets. Therefore the temporary hypothesis that the central action data were fundamentally nonholonomic was rejected.

8. Unreduced coefficient emitters

8.1 Complete Zhang-Kase Appendix-A 41-slot emitter

A complete ZK 41-slot emitter was reconstructed from action jets and frozen P5 geometry. It evaluates $a_1...a_9$, $b_1...b_5$, $c_1...c_6$, $d_1...d_4$, $e_1...e_4$ and $v_1...v_{13}$ using JET9D8 for radial derivative terms and the selected lower-order member when supplied.

Slot	median scaled rel	p95	max (edge sensitive)
a2	6.4e-09	2.18e-08	7.13e-06
c2	3.89e-11	3.03e-10	3.77e-10
d3	1.11e-07	7.39e-07	0.0363
e4	1.5e-06	1.72e-05	0.254
v1	6.95e-11	2.31e-10	2.89e-10
v4	1.15e-10	9.16e-10	1.52e-09
v6	5.55e-17	2.94e-16	6.47e-16
v12	1.39e-17	2.52e-16	6.56e-16
v13	6.83e-06	0.000154	1.5

The relatively large edge maxima for d3, e4 and v13 are consistent with known high-derivative edge sensitivity and archived rounded inputs; the interior median/p95 values are the meaningful regression measure.

8.2 Variable-G4(phi), G5=0 Maxwell-Horndeski emitter

The old strong-H principal action dataset was recovered with direct columns for G4(phi), G4_phi, G3_X, F=G=H, a1,a4,c4,mu and P1 over u approximately 0.551523...0.71. This made it possible to reconstruct the correct luminal variable-G4 Horndeski member rather than inventing it backward from a coefficient table.

Slot/quantity	median	p95	max
a1	1.6e-12	7.77e-12	2.26e-11
a2	2.42e-08	1.2e-07	3.53e-07
c2	0	0	0
c4	1.54e-12	8.91e-12	2.39e-11
c6	6e-13	3.73e-12	1.08e-11
d3	5.28e-11	3.84e-10	9.76e-08
e1	1.6e-07	6.31e-07	4.13e-06
e4	3.26e-08	1.43e-07	6.09e-07
mu	2.32e-12	1.37e-11	3.44e-11
K_scalar	4.62e-08	2.48e-07	1.46e-06

The Sep-16 parser fixes to c6, d3 and e4 reduce those expressions to numerical/JET-level regression. The emitter is therefore no longer an uncertainty in the final failure diagnosis.

9. Complete common even-parity constraint elimination

Before reduction the common gauge-fixed fields are (H0,H1,H2,h1,dphi,dA0,dA1,V). The physical basis is $\chi=(\psi,d\phi,V)^T$ with

$$\psi = H_2 + (L a_4/a_3) h_1 + (a_1/a_3) d\phi'.$$

Define $p=a_1/a_3$, $q=L a_4/a_3$ and

$$\begin{aligned} B &= a_2 - v_2 v_4/(2 v_1) \\ C\phi &= a_5 + L a_6 - v_2 v_5/(2 v_1) \\ CH_2 &= a_7 + L a_8 - v_2 v_3/(2 v_1) \end{aligned}$$

```

Ch1 = L (a9 - v2 v6/(2 v1))
Beff = B - a3 p' - CH2 p
Dh1 = Ch1 - a3 q' - CH2 q

```

$$h1 = -(a3 \psi' + \text{Beff} \, \text{dphi}' + \text{Cphi} \, \text{dphi} + \text{CH2} \, \psi + v2 \, V) / \text{Dh1}.$$

The remaining two algebraic time-derivative constraints use $\Delta V = 4 b1 v10 - v11^2$. Finally $\text{dA0} = (v1 \, V) / (L \, v9) - (v8 \, h1 + v12 \, H2 + v13 \, \text{dphi}) / (2 v9)$. The full JET9D8 implementation exports linear profile maps for $h1, H2, H1, \text{dA1}, \text{dA0}$ in $y = (\psi, \text{dphi}, V)$ and y' .

On the central 4000-point reference, $L=6$ gives $\min |\text{Dh1}| = 11.987940$, $\min |\Delta V| = 3.042884$ and $\min |2 L \, v9| = 0.119951$. The dphi'' and $h1'$ higher-derivative cancellations close to zero / about $1e-16$.

10. Profile-aware reducer and the missing S block

The decisive structural correction on Sep-16 was to retain the antisymmetric first-order radial block. Zhang-Kase Eq. 4.25 has the reduced form

$$L_{\text{even}}^2 = \text{dot}(Y)^T K \text{dot}(Y) + Y'^T G Y' + Y'^T S Y + Y^T M Y, \quad S^T = -S.$$

In the project convention used by the profile reducer, after applying the formal differential transformation T and integrating by parts:

```

K = -P20/2
G = +P02/2      (project action uses -chi'^T G chi')
R = -P11/2 = 0
S = 1/2 (P02' - P01)
M = -1/2 (P00 + S')

```

Earlier attempts to force the antisymmetric first-order radial contribution into M caused an apparent lower-order mismatch. Once S was retained explicitly, the reducer high- L mass regression passed.

10.1 Structural reducer audit

On the central reference there are no differential operator terms above second radial order, the $H0$ quadratic residual is about $4.55e-13$, and the internal principal K comparison is typically at $1e-12$ scale or better. The project has $R=0$ in the final profile convention.

11. High- L mass/potential closure

The direct-action high- L mass formulas serve as an independent regression target for the profile reducer. The corrected leading terms are:

```

M11 = -r^2 v6^2/(4 v1)
M13 = -r v6/2
M33 = -v1
M22 = e4 + 2 h c4 a6/a4 - m5_minus^2/(4 a4^2 v9)
z1 = (a4 + r a9 - 0.5 r A0prime v6) v6
m1_plus = +r a4 v8 + z1 # c4 sector
m1_minus = -r a4 v8 + z1 # dA0/v9 Schur sector
m2 = 2 c5 v1 + (A0prime v1 + 0.5 phiprime v4) v6
m3_plus = a4 v1' + 0.5 A0prime v1 v6
m4 = 2 A0prime v1 + phiprime v4
m5_minus = a4 v13 - a6 v6
M12 = -r d3/2 - h c4 z1/(a4 v6) - r h a6 m2/(2 a4 v1) + r v5 v6/(4 v1)
      - m1_minus m5_minus/(4 a4^2 v9)
      + (1/4) d/dr [2 r h c4 + r(2 d2 v1 - v4 v6)/(2 v1) + r v6 m5_minus/(2 a4 v9)]
M23 = h A0prime c4 v1/a4 + v5/2 + m3_plus m5_minus/(2 a4^2 v9)
      - h a6 m4/(2 a4) - (1/4) d/dr [v4 + v1 m5_minus/(a4 v9)]

```

The sign in $m5_minus$ and the distinction $m1_plus$ versus $m1_minus$ are essential. Spline-based mass/angular scans that violated these identities were superseded.

12. Why the old 41/41 tables could not simply be concatenated

A diagnostic global 41-stream concatenation was intentionally attempted and rejected. It exposed two separate issues:

1. The archived outer 37/41 table was a repaired pure-SVT branch, not the complete same-action Horndeski+SVT handover. Its coefficients became nonzero before the stored S_SVT partition actually turned on.

2. At $u=0.61$, the re-solved outer repaired-SVT endpoint and the Sep-13 central exact SVT table are different action-jet representatives on the same background. Their upper/background values f_3, f_4, N_4 are close, but f_3X, f_2, f_2X and Hessian jets are drastically different.

This is not a contradiction: background-constrained jets and background-null transverse jets are not unique. It does mean that a direct coefficient concatenation is not a proof of one same action.

Slot	outer at $u=.61$	central at $u=.61$	scaled jump
e2	-38.3087	56.5379	1.68
c2	-32.9208	54.7713	1.6
c4	4.50106	-10.8904	1.41
d2	3.79646	-9.34758	1.41
c6	7.74055	27.5645	0.719
d3	66.2971	20.1257	0.696
c5	-2.14589	-6.76487	0.683
a6	0.108629	0.669732	0.561
v13	94.8701	50.8563	0.464
e4	375.383	206.902	0.449

Interpretation

The existing central_outer_matching_buffer in the global principal export is a characteristic/principal-basis matching section. It does not by itself certify a finite-I action-level 41-slot handover between two different transverse representatives.

13. Sep-16 outer finite-I endgame - what was tried and what each attempt taught us

13.1 First simple cubic/Einstein-like MH member

A deliberately simple $G_4=1/2, G_5=0$ Maxwell-Horndeski member was combined with the genuine-SVT Delta-C block. Its structural reduction was finite, but K became negative. Large scans of the genuine-SVT Hessian controls, including Delta v1 variations far beyond the initial value, barely moved the bad kinetic direction. The problematic eigenvector was nearly scalar. Therefore the failure was not fixable by simply turning the radial f_2 Hessian knobs.

13.2 Control experiment with the archived strong-H carrier

The same genuine-SVT Delta-C contribution combined at coefficient level with the already healthy strong-H principal carrier gave a high-L K with no negative rows in the diagnostic $L=1000$ run (minimum eigenvalue approximately $1.18e-6$). This isolated the missing freedom to the Horndeski principal member, not the ZK emitter or the reducer.

13.3 Recovery of the actual strong-H action data

The archived file `ssz_p5_horndeski_carrier_through_light_rings_to_core_2026-09-12.csv` was recovered. It directly stores the variable $G_4(\phi)$, G_4_ϕ and G_3_X principal carrier. The strong-H table has $F=G=H$ and a positive target scalar kinetic margin about $9.7423e-3$. This was the correct input for a variable- $G_4(\phi)$, $G_5=0$ MH emitter.

13.4 Auxiliary-V recanonicalization after hybrid assembly

A major algebraic trap was discovered: adding two coefficient sets that had each already been auxiliary-V completed does not preserve the H_0^2 identity. After summing the common unreduced blocks, the shared auxiliary V sector must be canonicalized once again. The required identity is $v_7=v_2^2/(4v_1)$ in the corresponding canonical route. A pre-recanonicalization residual around 0.684 was therefore an assembly error, not physics. After common recanonicalization, the residual is exactly zero in the final audit.

13.5 Final variable- G_4 outer candidate - important on-shell caveat

The final outer candidate used:

- the variable-G4(ϕ), $G_5=0$ healthy MH principal carrier reconstructed from the archived strong-H action data;
- the re-solved outer geometry and A_0' profile;
- the genuine-SVT Delta-C block assembled before constraint elimination;
- common auxiliary-V recanonicalization after assembly;
- $v_{12}=-v_6/(2h)$;
- selected lower-order member $v_5=c_3=e_3=0$;
- a_5 regenerated holonomically with JET9D8.

All 41 slots are finite on the 2201-point outer grid, and several assembly identities close. However, this construction is not a strict certified same-action/on-shell member: the variable-G4 strong-H principal data were interpolated onto the already re-solved outer background that had been constructed for a different Horndeski representative, and the full background equations were not re-solved after that substitution. The file names containing FINAL_SAME_ACTION therefore overstate the status and should be read as historical run names. The candidate is nevertheless rejected because the symmetric kinetic quadratic form is negative over a large radial set.

14. Final variable-G4 candidate operator audit - decisive for the candidate, not a theory no-go

For this candidate, the symmetric part of the 3×3 kinetic quadratic form is not positive definite. The audit code explicitly diagonalizes $K_s=(K+K^T)/2$. This is the correct quadratic-form test even when a small numerical antisymmetric remainder is present. The detailed audit across L is:

L	min eig(K)	negative rows	min radial c^2	negative c^2 rows	max S sym residual	max R
6	-2.116849e+03	2113	8.882930e-02	0	2.53e-08	0.0e+00
12	-6.688485e+02	2065	9.061206e-02	0	2.08e-08	0.0e+00
20	-2.363717e+02	2025	6.387482e-02	0	1.51e-08	0.0e+00
42	-1.286435e+01	1903	2.852613e-03	0	5.24e-09	0.0e+00
110	-1.884584e+00	1576	9.503906e-04	0	1.08e-08	0.0e+00
420	-2.505590e-01	1061	2.541722e-04	0	9.09e-09	0.0e+00
1000	-1.064799e-01	629	1.075190e-04	0	1.71e-08	0.0e+00

Additional checks: the H_0 auxiliary identity and v_{12} identity close exactly, no forbidden higher-order operator keys appear, $R=0$, G is symmetric to floating-point accuracy, and S is antisymmetric at about $1e-8$. The raw time-principal block P_{20}/K , however, retains a nonzero antisymmetric numerical remainder in this transplanted candidate (for the stored $L=6$ and $L=42$ profiles the relative max asymmetry is about $1.6e-4$; at $L=1000$ about $1.2e-6$). This does not affect the quadratic form after symmetrization, but it is another reason not to label the candidate a fully clean final member. Constraint pivots remain nonzero.

14.1 Spatial location of the kinetic failure

Using the stored $L=6,42,1000$ matrix profiles, the negative region begins progressively farther inward as L increases:

L	negative-K interval on stored grid	location of most negative point
6	u about 0.552613 to 0.610000	minimum near $u=0.608139$
42	u about 0.554474 to 0.610000	minimum near $u=0.608139$
1000	u about 0.566913 to 0.600830	minimum near $u=0.583366$

The high- L weakening explains why principal-only or asymptotic diagnostics can look healthy while finite- I remains unacceptable. It also shows why QNM must wait for an explicitly finite- I healthy member.

14.2 What the failure does and does not mean

Does mean

The Sep-16 variable-G4 transplant candidate must be rejected. Its symmetrized kinetic quadratic form has a negative direction over a large part of the outer band, and it was not independently re-solved on the background after changing the Horndeski principal member.

Does not mean

It does not show that a certified global same-action finite-I member has failed, because this candidate was not fully background re-solved after the variable-G4 substitution. It also does not invalidate the P5 geometry, local strong-H witness, central genuine-SVT light-ring witness, constructive same-action existence theorem, 41/41 coefficient completion, derivative service, constraint maps, ZK emitter, MH emitter, or profile reducer.

15. QNM status and why calculation stops here

A coupled QNM spectrum is not scientifically meaningful at this stage. The last candidate is rejected and no certified center-to-infinity on-shell finite-I member has yet passed the gates. The QNM gate is therefore deliberately closed.

Independent work on the asymptotic exterior remains useful: an outgoing Jost expansion was transported with order-10 error around $2.43e-12$. The previously attempted inward shooting of damped modes on the real radial axis was rejected because the growing complementary solution contaminates the integration and candidate roots moved when the outer radius changed.

Acceptable production approaches after a healthy global KRGm member exists include exterior complex scaling, compactified Jost/spectral determinants, or a continued-fraction-type formulation. Any eventual root must be tested for radial resolution, domain/outer-boundary, complex-scaling angle, basis normalization, branch tracking and determinant convergence.

16. Superseded and rejected routes - do not accidentally reuse

Route	Status	Reason
$v_{12}=+v_6/(2h)$	SUPERSEDED	Accepted Appendix branch is $v_{12}=-v_6/(2h)$.
a5 without -a1"	SUPERSEDED	Correct a5 includes -a1 double prime and vector terms.
Spline-based negative mass/angular scans	SUPERSEDED	They violated later action-level high-L identities.
Central member fundamentally nonholonomic	REJECTED	Direct chain-rule tests close with JET9D8.
A2 requires +4h instead of +2h	REJECTED	Came from a manual derivative error.
Naive inner E_H+E_{SVTg} residual is the full residual	SUPERSEDED	Re-solved handover includes omitted partition and lower-jet terms.
Eq. A2/131 repair member as production member	NOT ACCEPTED	Diagnostic/existence probe only.
Old F3 inner 41/41 coefficient blend	CANDIDATE ONLY	Pending direct action-level Appendix re-evaluation.
Old outer repaired-SVT 37/41 as full handover	REJECTED FOR HANDOVER	It is a repaired pure-SVT branch, not the same-action partition.
Simple $G_4=1/2$ outer candidate	REJECTED	Too restrictive; produced negative scalar kinetic channel.
Variable-G4 transplanted outer candidate	REJECTED CANDIDATE	Negative sym(K); principal member was transplanted without a fresh full background re-solve.
Real-axis inward QNM shooting	REJECTED	Numerically contaminated for damped modes.

17. Correct future continuation - one targeted control problem, not another tuning loop

If the project is resumed, the next task should not be another hand-picked representative. The correct problem is to formulate the remaining Horndeski/SVT transverse jet freedom as a constrained continuation/optimization problem with the finite-I kinetic gate included explicitly.

1. Freeze the already accepted geometry, partition functions, common basis, v_{12} sign, lower-order convention, JET9D8 service and reducer.
2. Parameterize only genuinely background-null / representative-selection directions. Do not alter jets already fixed by the frozen background equations unless they are re-solved consistently.
3. Use the full-rank Horndeski control family, not the over-restricted $G_4=\text{constant}$ subclass. Nonredundant principal targets are (F,H,a1,c4,c2).

4. After selecting a new Horndeski principal member, re-solve the complete static background equations for the combined HSVT action before evaluating finite-I stability. Do not transplant principal jets onto a background solved for another member.
5. Enforce exact background residuals and the common constraint pivots simultaneously with a finite-I lower bound $K \geq \epsilon$ on selected L gates, beginning with $L=6$ and $L=42$.
6. Only after $K>0$, solve/check radial generalized eigenvalues and the angular large-L symbol. Then regenerate all 41 coefficients directly from the chosen action jets.
7. Build the action-level Outer->Central and Central->Inner matching sections; do not reuse the principal matching buffers as proof of finite-I action continuity.
8. Freeze one global 41/41 stream and rerun the complete profile reducer. Only then open the QNM gate.

The previously suggested 00/11 inversion/rotation idea remains a useful diagnostic only where the background inverse chart changes conditioning. The mathematically justified form is the E00, E11-E00 basis associated with $\kappa=-2X$. It is not a substitute for kinetic stability control.

18. Reproducibility cookbook

18.1 Authoritative scripts as of this checkpoint

File	Role
ssz_p5_higher_jet_closure_2026-09-16.py	JET9D8 nonuniform radial derivative/jet service.
ssz_p5_holonomic_a5_closure_2026-09-16.py	Accepted a5 holonomic reconstruction.
ssz_hybrid_full_constraint_maps_JET9D8(1).py	Complete algebraic maps for h1,H2,H1,dA1,dA0.
ssz_hybrid_unreduced_even_kernel.py	Common unreduced action architecture / canonical slot basis.
ssz_hybrid_jet_aware_constraint_reducer.py	First-stage generalized-psi reducer.
ssz_p5_profile_operator_reducer_JET9D8_2026-09-16.py	Formal profile-aware $T^{\dagger} P T$ reducer producing K,G,S,M.
ssz_p5_F2b_action_derived_highL_mass_closure_2026-09-15.py	Independent high-L mass identities/regression.
ssz_p5_zk_appendixA_emitter_JET9D8_2026-09-16.py	Complete Zhang-Kase Appendix-A 41-slot emitter.
ssz_p5_mh_luminal_g4phi_emitter_JET9D8_2026-09-16.py	Variable-G4(phi), G5=0 Maxwell-Horndeski emitter after parser fixes.
build_outer_resolved_zk_delta_2026-09-16.py	Outer resolved ZK raw and genuine-SVT Delta-C construction.
final_outer_variableG4_2026-09-16.py	Final Sep-16 variable-G4 transplant candidate assembly/audit; historical naming overstates same-action status.

18.2 Critical data products

File	Meaning
ssz_p5_integrable_svt_unreduced_even_coefficients_2026-09-12.csv	Central raw 41-slot ZK coefficient source.
ssz_p5_integrable_full_svt_lobe_ZK_exact_regression_2026-09-13.csv	Central exact principal/stability regression.
ssz_p5_horndeski_carrier_through_light_rings_to_core_2026-09-12.csv	Recovered strong-H action/principal carrier including G4,G4_phi,G3X.
ssz_p5_F2_horndeski_carrier_unreduced_39of41_CORRECTED_2026-09-14.csv	Archived strong-H coefficient regression target.
ssz_p5_F2_outer_same_action_RESOLVED_background_jets_2026-09-15.csv	Machine-zero outer background handover action jets and partition.
ssz_p5_F2_inner_same_action_RESOLVED_v2_background_jets_2026-09-15.csv	Machine-zero inner background handover action jets.
ssz_p5_F2A_GLOBAL_CANONICAL_KRG_PRINCIPAL_CINF_FINAL_2026-09-15(2).csv	45,617-point global principal K/R/G witness.
ssz_p5_OUTER_GENUINE_SVT_DELTA_41of41_2026-09-16.csv	Outer genuine-SVT beyond-shared coefficient contribution.
ssz_p5_MH_G4PHI_EMITTER_STRONGH_REGRESSION_2026-09-16.csv	Final MH emitter regression statistics.
ssz_p5_ZK_APPENDIX_A_EMITTER_OUTER_REGRESSION_2026-09-	ZK emitter regression statistics.

File	Meaning
16.csv	
ssz_p5_OUTER_FINAL_SAME_ACTION_41of41_2026-09-16.csv	Historical filename: variable-G4 transplant candidate, not a freshly background-re-solved same-action member; candidate fails sym(K).
ssz_p5_OUTER_FINAL_SAME_ACTION_OPERATOR_AUDIT_2026-09-16.csv	Finite-I audit of the rejected transplant candidate; decisive only for that candidate.
SSZ_P5_FINAL_STOP_CHECKPOINT_2026-09-16.md	Short frozen stop-state checkpoint.

18.3 Minimal resume sequence

1. Read SSZ_P5_FINAL_STOP_CHECKPOINT_2026-09-16.md
2. Read this PDF, especially Sections 2, 10, 13-17
3. Do NOT regenerate 41/41 from scratch
4. Do NOT revert v12 sign or a5
5. Do NOT drop S
6. Do NOT add reduced operators sector-by-sector
7. Start from final_outer_variableG4_2026-09-16.py and replace only the representative-selection/control layer
8. Impose $K>0$ at finite L as a target during the action-jet continuation
9. Re-solve background-constrained jets
10. Re-run full profile reducer and all acceptance gates
11. Only after PASS: construct coupled QNM determinant

19. Literature and convention anchors

The two perturbation formalisms used throughout the coefficient work are:

- C. Zhang and R. Kase, Physical Review D 110, 044047 (2024), arXiv:2404.11910 - even-parity perturbations in U(1)-gauge-invariant scalar-vector-tensor theories; Appendix-A coefficients and reduced K/G/S/M structure.
- R. Kase and S. Tsujikawa, Physical Review D 107, 104045 (2023), arXiv:2301.10362 - odd/even black-hole perturbations in Maxwell-Horndeski theories; full five-degree-of-freedom static-spherical formalism.

Project conventions that must stay locked when comparing formulas: $r_s=1$; $u=C=r_s/r$; $x=r/r_s=1/u$; scalar $\phi=X_i$; the accepted vector numbering uses the 13-slot Zhang-Kase schema after canonicalizing the Maxwell-Horndeski $v_1\dots v_9$ vector block; the selected lower-order member used in the final outer run has $v_5=c_3=e_3=0$.

20. Frozen final statement

What can safely be said as of 16 September 2026

P5 has a regular frozen geometry, healthy local principal witnesses in the exterior, genuine-SVT light-ring lobe and core, constructive same-action handover existence, a complete common 41-slot unreduced coefficient language, a validated derivative service, complete constraint maps, validated Zhang-Kase and Maxwell-Horndeski coefficient emitters, and a profile-aware finite-I reducer including the antisymmetric S term. No certified global finite-I same-action member has yet been completed. The last variable-G4 outer candidate is rejected because sym(K) is negative and, more fundamentally, the changed Horndeski principal member was not followed by a fresh full background re-solve. Therefore coupled QNM remains blocked. Future work should formulate a targeted transverse action-jet continuation/control problem with background equations and finite-I K positivity imposed simultaneously.

This report intentionally records both successes and the final failure so that a later continuation does not accidentally erase hard-won distinctions or redo already closed algebra.

Appendix A. Exact final operator-audit rows

A.1 Physical kinetic/radial gate and constraint pivots

L	min eig K	neg K	min c^2	neg c^2	min Dh1	min DeltaV	min 2Lv9
6	-2116.84928	2113	0.08882930162	0	3.387397372	0.1871260537	12.87893151
12	-668.8484547	2065	0.09061205538	0	14.11018473	0.1871260537	25.75786302
20	-236.3716979	2025	0.06387482154	0	39.81784118	0.1871260537	42.92977171

L	min eig K	neg K	min c^2	neg c^2	min Dh1	min DeltaV	min 2Lv9
42	-12.86435355	1903	0.002852612567	0	177.7549713	0.1871260537	90.15252058
110	-1.884583715	1576	0.000950390561	0	1227.614249	0.1871260537	236.1137444
420	-0.2505590114	1061	0.000254172238 4	0	17952.08463	0.1871260537	901.5252058
1000	-0.1064798939	629	0.000107519043	0	101833.7002	0.1871260537	2146.488585

A.2 Structural residuals of the same final runs

L	H0 quadratic	P20 asym	P02 asym	P11 asym	S sym residual	max R	v12 residual
6	0	1.43	1.78e-15	0	2.53e-08	0	0
12	0	0.521	8.88e-16	0	2.08e-08	0	0
20	0	0.229	8.88e-16	0	1.51e-08	0	0
42	0	0.063	4.44e-16	0	5.24e-09	0	0
110	0	0.0104	2.22e-16	0	1.08e-08	0	0
420	0	0.000763	5.55e-17	0	9.09e-09	0	0
1000	0	0.000136	2.78e-17	0	1.71e-08	0	0

Appendix B. Final emitter regression highlights

Zhang-Kase outer regression: selected median scaled-relative values.

slot	median	p95	max scaled	max abs
a2	6.3971216e-09	2.1848453e-08	7.13428e-06	7.13428e-06
a7	1.5325416e-09	7.9848129e-09	2.9541308e-06	4.1852845e-06
c2	3.8932921e-11	3.0284995e-10	3.7652955e-10	9.9212407e-09
c5	5.6290558e-10	1.4581929e-09	3.2625389e-07	5.4396854e-07
c6	6.4394103e-10	2.1054491e-09	5.7237983e-07	8.9151975e-07
d3	1.1077117e-07	7.3913578e-07	0.036320295	0.096146351
e1	9.2607537e-09	1.624784e-07	6.9497323e-05	6.9497323e-05
e4	1.5017315e-06	1.7209745e-05	0.25374308	24.009088
v1	6.9516516e-11	2.3093623e-10	2.8907549e-10	3.4672958e-09
v4	1.1493828e-10	9.1617842e-10	1.5155912e-09	1.5155912e-09
v9	5.3223806e-07	1.8204314e-06	1.3474865e-05	2.4951214e-05
v13	6.8309907e-06	0.00015434887	1.5038073	13.171992

Maxwell-Horndeski variable-G4 strong-H regression: selected values.

slot/quantity	median	p95	max
a1	1.5989224e-12	7.7720018e-12	2.2622793e-11
a2	2.4183403e-08	1.1965377e-07	3.5311115e-07
a7	1.0504046e-08	6.001481e-08	3.8532134e-07
b3	2.0813369e-10	1.7727104e-09	9.834189e-08
c2	0	0	0
c6	6.0049188e-13	3.7273101e-12	1.0817791e-11
d3	5.2794574e-11	3.8395302e-10	9.7565025e-08

slot/quantity	median	p95	max
e1	1.6026991e-07	6.3109237e-07	4.1328654e-06
e4	3.2637367e-08	1.429779e-07	6.0942887e-07
mu	2.3217498e-12	1.3672322e-11	3.4381303e-11
K_scalar	4.6206209e-08	2.4838841e-07	1.4557856e-06

Appendix C. Compact decision log

Stage	Finding	Rule for future
Before Sep-16	41/41 local selected tables existed	Keep: slot closure was real.
Sep-16 early	Profile reducer initially treated lower radial first-order term incorrectly	Fixed by explicit antisymmetric S block.
Sep-16	ZK Appendix-A emitter reconstructed	Keep and reuse.
Sep-16	Outer repaired-SVT endpoint found not identical to central Sep-13 action representative	Do not concatenate as same action.
Sep-16	Simple G4=1/2 MH outer member gave negative K	Reject that restricted member only.
Sep-16	Hessian scans failed to move bad scalar kinetic direction	Do not waste time on f2 Hessian-only tuning.
Sep-16	Strong-H diagnostic restored positive high-L K	Horndeski principal freedom is relevant.
Sep-16	Recovered direct variable-G4 strong-H action data	Use as MH action anchor.
Sep-16	c6,d3,e4 parser/holonomic fixes completed	MH emitter regression closed.
Sep-16	Auxiliary-V block found non-additive after separate completion	Always recanonicalize common V after assembly.
Sep-16 final	Variable-G4 principal carrier transplanted into resolved outer grid and reduced	Rejected candidate: negative sym(K), nonzero raw K asymmetry, and no fresh full background re-solve after transplant.
Stop rule	No more ad-hoc tuning in this run	Resume only as constrained control/optimization problem.

Generated from the frozen project artifacts present in the working archive on 16 September 2026. Numerical values in this report are intended as reproducibility anchors; the CSV and Python files listed in Section 18 remain the machine-readable source of truth.